EL MEHDI HARESS

PhD in Applied Mathematics | Teacher and researcher in probability and statistics

Probability | Statistics | Stochastic modelling | Quantitative finance | Data-driven decisions

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Sophia Antipolis, France — willing to relocate to Lyon



PROFILE

Applied mathematician (PhD, Université Paris-Saclay; engineer, CentraleSupélec) with **five years of university teaching** in probability, statistics, PDEs and stochastic finance, from first-year undergraduates to M2, in **French and English**, including students outside mathematics (economics, engineering). I teach mathematics through the questions it answers: real data sets, simulation in Python, and student projects on real-world problems. Research in stochastic modelling, statistical inference and risk-aware learning (8 peer-reviewed papers, including 2 at *AISTATS*), with ongoing work alongside industrial partners at Inria.

TEACHING EXPERIENCE

09/2026 — now
Lecturer — Statistics for non-specialists (L2)
Université Côte d'Azur, Nice
- Statistics (L2) for economics undergraduates, with course notes written for the module.
- Supervision of a project for second-year engineering students (Polytech Nice): analysis of real wind and temperature time series from an atmospheric observatory (SIRTA).

Fall 2025
Teaching assistant — Stochastic Processes (undergraduate)
University of Leeds (UK) — in English
- Weekly problem-solving workshops, 50 students.

2021 — 2024
Teaching assistant (about 60 students per year)
CentraleSupélec, Paris-Saclay
- Convergence, Integration and Probability (2022–2024); Partial Differential Equations (2021–2024).
- **Stochastic Finance and Risk Modelling** (postgraduate, 25 students, 2022–2023).

2017 — 2024
Tutoring, examining and coordination
CentraleSupélec, Université Paris-Saclay, Lycée Chaptal (Paris)
- Oral examiner in mathematics for preparatory classes (Lycée Chaptal, 2022–2024).
- Tutor for first-year mathematics students (CentraleSupélec, 2017–2023); coordinator of the tutors of the Paris-Saclay mathematics support programme (2023).

RESEARCH AND ACADEMIC POSITIONS

09/2026 — now
Research engineer — probabilistic numerical methods
Inria Sophia Antipolis — CALISTO team (Mireille Bossy)
- Lagrangian stochastic models for turbulence and multi-scale systems from industrial applications; work with engineers and industrial partners.

12/2024 — 08/2026
Research fellow in stochastic analysis
University of Leeds (UK) — K. Dareiotis and K. Lê
- Stability and long-time behaviour of singular stochastic dynamics (*SPA*, 2026).

10/2021 — 11/2024
PhD — numerical approximation and long-time behaviour of singular SDEs and SPDEs
Université Paris-Saclay / CentraleSupélec — L. Goudenège and A. Richard
- Simulation schemes with proven convergence; estimation of several parameters from discrete data.

09/2019 — 09/2021
Research internships
Osaka University (Japan) — M. Holland; University of Alberta (Canada) — Y. Hu
- Learning under risk constraints with heavy-tailed data (2 *AISTATS* papers); parameter estimation in a fractional Ornstein–Uhlenbeck model.

EDUCATION

2021--2024 PhD in Applied Mathematics — Université Paris-Saclay.
2020--2021 M2 Mathematics of Randomness (probability and statistics) — Université Paris-Saclay.
2017--2021 Engineering degree, Applied Mathematics — CentraleSupélec.
2015--2017 CPGE (MP, MP*) — Lycée Pierre de Fermat, Toulouse.

PUBLICATIONS

Author order is alphabetical in probability.

- K. Dareiotis, **E.M. Haress**, K. Lê. *Uniform pathwise stability of singular additive SDEs driven by fractional Brownian motion*. Stochastic Processes and their Applications, 2026.
- L. Goudenège, **E.M. Haress**, A. Richard. *Numerical approximation of the stochastic heat equation with a distributional reaction term*. IMA Journal of Numerical Analysis, 2026.
- L. Goudenège, **E.M. Haress**, A. Richard. *Numerical approximation of stochastic differential equations with distributional drift*. Stochastic Processes and their Applications, 2025.
- **E.M. Haress**, A. Richard. *Estimation of several parameters in discretely-observed stochastic differential equations with additive fractional noise*. Statistical Inference for Stochastic Processes, 2024.
- **E.M. Haress**, A. Richard. *Long time Hurst regularity of fractional SDEs and their ergodic means*. Journal of Theoretical Probability, 2025.
- M. Holland, **E.M. Haress**. *Spectral risk-based learning using unbounded losses*. AISTATS, 2022.
- M. Holland, **E.M. Haress**. *Learning with risk-averse feedback under potentially heavy tails*. AISTATS, 2021.
- **E.M. Haress**, Y. Hu. *Estimation of all parameters in the fractional Ornstein–Uhlenbeck model under discrete observations*. Statistical Inference for Stochastic Processes, 2021.

Submitted and in progress: K. Dareiotis, **E.M. Haress**, K. Lê. *Regularising flows and singular fractional SDEs with superlinear dissipation*, preprint (2026). • L. Goudenège, **E.M. Haress**, J. Naffrichoux. *Singular perturbation of dissipative SDEs*. • **E.M. Haress**, A. Mayorcas, D. O'Kane. *Replicating Gibbs measures by invariant measures of fBm-driven SDEs*. • Lagrangian stochastic models for turbulent flows (with M. Bossy, Inria).

SERVICE AND ENGAGEMENT

Organisation	Co-organiser of FRSSD 2026, international workshop, Leeds (18 speakers: programme, website, budget). Co-organiser of CJC-MA 2023, national conference (143 participants).
Collaboration	Weekly research collaboration with a PhD student (numerical methods for SPDEs, since 2025). Member of the French team of the ECOS-Sud project C25E10 with Universidad de Concepción (Chile).
Community	Reviewer for *SPA*, *J. Theoretical Probability* and *J. Computational and Applied Mathematics*; editorial coordination of the CJC-MA 2023 proceedings (*ESAIM: Proceedings and Surveys*). Member of COST Action CA24104 Stochastica. 15+ talks, including invited seminars and talks at Warwick, Rennes (IRMAR), Évry and *Les Probabilités de Demain*.
Distinctions	Vivaldi programme, Fondation Mathématique Jacques Hadamard (2023); IDEX international mobility grant, Paris-Saclay (2019).

SKILLS

Teaching tools	**Python** (NumPy, SciPy, pandas) — 8 years, 5 public research repositories; R, MATLAB; LaTeX course notes; web apps built end to end (React, TypeScript, Firebase), usable for interactive teaching material.
Languages	French — native. English — fluent (2 years teaching and research in the UK). Japanese — beginner.

REFERENCES

L. Goudenège	Research Director (DR), CNRS, Laboratoire de Mathématiques et Modélisation d'Évry (LaMME) — PhD supervisor. ludovic.goudenege@univ-evry.fr
M. Bossy	Inria, Centre d'Université Côte d'Azur, CALISTO team — current supervisor. mireille.bossy@inria.fr